Hardware

Working upload verified

One Arduino UNO Q reads the office and data-center models, drives their local outputs, and reports both buildings in one HTTPS snapshot.

Hardware and wiring

Pin map from the current sketch

Signal	Office	Data center
Temperature	KY-028 analog A1, digital D2	KY-028 analog A5, digital D3
Light or flame/IR	Grove light on A0	Flame/IR digital output on D4
Model fan	Servo on D9	Servo on D10
Indicator LEDs	White on D5, D6, D11	Green on A2, A3, A4

The sketch initializes all six LEDs on. Both servo fan models are configured to sweep by default, changing target between 10 and 170 degrees every 500 ms. The reported fan state represents the controller's movement request. RPM and physical movement have no feedback.

Electrical and bench checks

Use the UNO Q maker-header limits when checking module supply and output voltage. Its analog reference is 3.3 V by default. Use current-limiting resistors on LEDs, a suitable external servo supply with common ground, and verify voltage and repeated movement on the actual assembly.

The sketch expects HIGH on the flame input and LOW on the thermistor digital hot input. Confirm both polarities at the bench. Record the exact flame module and document what causes an activation.

Arduino UNO Q power and I/O specifications

Telemetry and reference data

HARDWARE

The STM32 sketch reads 10-bit ADC counts from 0 to 1023 and exposes a snapshot through Arduino Bridge. The Linux Python uploader reads that snapshot, timestamps it in UTC, and sends it about once per second, plus request time. An activation counter preserves brief IR events until a current upload succeeds.

Field group	Data sent
Snapshot envelope	Schema v1, unoq-main, boot ID, sequence, MCU uptime, UTC capture time, hardware source.
Office	Relative light count, thermistor count and heat flag, controller LED and fan state.
Data center	Raw flame digital level, active and brief-activation flags, thermistor count and heat flag, controller LED and fan state.

Production sample · Nov 8, 2026, 8:44 a.m. Central

Building	Observed snapshot	Approx. temperature
Office	Light 545/1023, temperature 243 counts, heat flag active, fan running.	21.1 °C / 70.0 °F
Data center	Flame input low and clear, temperature 390 counts, heat flag inactive, fan running.	22.5 °C / 72.5 °F

The room display read 71.7 °F. The respective startup references were 235 and 395 counts. The backend estimates degrees using one reference per sensor and shows raw counts beyond 10 °C from it. The separate heat flag uses the startup count or module digital output. It is not an absolute temperature alarm.

KY-028 behavior and module documentation

Bring-up and acceptance

HARDWARE

Start the UNO Q app

01 Connect the board to Wi-Fi in Arduino App Lab or from its Linux desktop. Confirm the board can reach the hosted /api/config over HTTPS. A different network from the dashboard is fine.

02 Copy the repo sketch and Python uploader into the App Lab app. Create python/device.json next to main.py with the hosted backend_url and the private device_token. Select Run and check for Sent reading.

03 Open Sensor Data with the owner account. Confirm the office light, both temperatures, the data-center IR status, and both fan states match the controller observations. Use Serial Monitor for raw counts and flags.

Remaining checks for the hardware handoff

01 Log repeated normal and reduced office light samples while the LEDs remain reported on. Provide the two count ranges for the production detection threshold.

02 Confirm actual flame module polarity and exercise both a brief IR pulse and a sustained activation with a controlled bench test. Save the corresponding snapshots.

03 Stop the uploader and confirm that both buildings become stale. Restart it and confirm that reporting resumes with a fresh boot ID and sequence. Save a short capture with no credentials.

04 Keep local indicator and fan behavior as shipped for the demo. Remote output commands require a firmware or uploader handler and a physical bench test before they enter the MVP acceptance criteria.

UNO Q App Lab setup and calibration

Hosted BuildingConnect dashboard